

PAVAN KUMAR SADASHIV

AI Systems Chief Architect | GPU Kernel & High-Performance Runtime Engineer | Founder & MD

Mangaluru, Karnataka, India | Email: hrlinternationalprivatelimited@gmail.com | GitHub: github.com/hrlpavan
Platform: hrlpavan.github.io/hrl-international-website- | Charter: github.com/hrlpavan/engineering-compensation-blueprint

PROFESSIONAL SUMMARY

High-velocity AI Systems Chief Architect and GPU Systems Engineer with specialized expertise in low-level CUDA/C++20 high-performance inference runtimes, Paged KV-Cache virtualization, CUDA Graph optimization, hardware-aware FP8/INT4-AWQ quantization, and discrete combinatorial optimization (NP-hard CSP Backtracking). Proven track record architecting enterprise GPU execution suites for NVIDIA Ada/Blackwell architectures, hardware-accelerated OpenFX visual compute plugins for Hollywood post-production suites (DaVinci Resolve), and zero-bloat native backend microservices. Founder and Managing Director of HRL International Private Limited. Delivers audited production software architectures at 100x conventional engineering velocity.

CORE TECHNICAL COMPETENCIES

GPU & Low-Level Systems:	C++20/17, NVIDIA CUDA 12.6, Apple Metal (MSL), CUDA Graphs, Paged Attention & KV-Cache, OpenFX API, GPU Memory Optimization
AI/ML & High-Performance Inference:	PyTorch, ONNX Runtime, FP8 (E4M3/E5M2) & INT4-AWQ Quantization, Speculative Decoding, Depth Anything AI, ACESc Color Science
Algorithms & Discrete Optimization:	Combinatorial Optimization, NP-Hard Constraint Satisfaction (CSP) Backtracking, Dynamic K-Means (k=3), Bayesian Classification
Backend & Systems Architecture:	Native Node.js (Zero-Framework Architecture), SQLite 3 (ACID Transactions), REST APIs, Google Antigravity SDK, Cloud Composer, BigQuery
DevOps & Build Systems:	CMake, Git, GitHub Actions (CI/CD), macOS DMG Application Bundling, Linux Kernel/Driver Environments

FLAGSHIP ENGINEERING & GPU PROJECTS

1. NVIDIA RTX-LocalAI Enterprise Execution Suite

Technologies: C++20, CUDA 12.6, Paged KV-Cache, CUDA Graphs, Speculative Decoding, INT4-AWQ, FP8, CMake | hrlpavan.github.io/nvidia-project-by-hrl/

- Architected a C++20/CUDA 12.6 on-device AI inference engine and memory management subsystem for NVIDIA Ada Lovelace / Blackwell & DGX systems.
- Paged KV-Cache Virtualization:** Engineered a block manager (16 tokens/block) achieving **0.17 µs/request latency** with **0.00% memory fragmentation** on 8GB–24GB VRAM.
- CUDA Graphs & Continuous Batching:** Pre-captured execution graphs dropping CPU launch latency to **< 5 µs** with iteration-level continuous batching (512 tokens/chunk).
- Speculative Policy Engine & FP8/INT4 Kernels:** Implemented draft-target verification loops for **up to 2.2x speedup** (88.4% acceptance) and custom dequant GEMM/GEMV CUDA kernels.

2. AI Cinematic Haze — OpenFX & Fusion GPU Plugin for DaVinci Resolve

Technologies: C++17/20, NVIDIA CUDA, Apple Metal (MSL), OpenFX API, Depth Anything AI, ONNX Runtime, ACESc | github.com/hrlpavan/hrl-international-website-

- Architected a high-throughput OpenFX visual compute plugin with native CUDA and Apple Metal backends for DaVinci Resolve 21.
- Integrated monocular neural depth estimation (Depth Anything via ONNX Runtime) to compute real-time volumetric optical atmospheric scattering.
- Preserved uncompressed 12-bit DPX sensor dynamic range through strict Hollywood ACESc color science management.

3. InviMatrix — College Examination & Invigilation Management Portal

Technologies: JavaScript (ES6+), Native Node.js (Zero-Framework), SQLite 3, CSP Backtracking, K-Means Clustering | github.com/hrlpavan/exam-invigilation-dsa-app

- Implemented an NP-hard Constraint Satisfaction Problem (CSP) Backtracking solver eliminating scheduling collisions across faculty and venues.
- Built an AI/ML Smart Batch Planner with dynamic K-Means (k=3) & Bayesian classifier for anti-cheating student seat dispersion.
- Engineered a zero-framework native Node.js REST server backed by SQLite with strict ACID-compliant transactions.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Founder & Managing Director — HRL International Private Limited (2026 – Present)

- Founded enterprise deep-tech software venture; authored Startup India Seed Fund & Karnataka ELEVATE grant dossiers.
- Author of *Rule Breaking (Volume 4.0): The 100x AI Chief Architect Manifesto* (Master Treatise).
- Architected automated media distribution infrastructures generating 2.5M+ organic impressions across digital creator channels.

EDUCATION & VERIFIED CERTIFICATIONS

Bachelor of Engineering (B.E.) in Computer Science & Engineering (AI & ML)

Sahyadri College of Engineering & Management (SCEM), Mangaluru, Karnataka, India

- **JPMorgan Chase & Co.** — Distributed Streaming & Software Engineering Simulation (Kafka, REST Controller Architecture)
- **Deloitte Australia** — Cybersecurity Defense & Enterprise Threat Modeling Simulation (Perimeter Defense, Incident Response)
- **Blackmagic Design** — DaVinci Resolve Color Science & Visual Computing Integration